Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 17)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 A conservation team is planning to protect a population of an endangered antelope in a reserve.

0 1 . 1 Explain why each of the following features of the population should be measured before a conservation strategy is decided.

[8 marks]

Population size .....

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Distribution .....

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Survival rate .....

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Age structure .....

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0 1 . 2 Describe how the population size of the antelope could be estimated.

[2 marks]

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0 2 . 1 Describe the ecological features of mangroves and the adaptations of mangrove trees.

[3 marks]

0 2 . 2 Explain **four** reasons why mangroves are important.

[4 marks]

1

2

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0 2 . 3 Describe the main threats to mangroves and the conservation efforts used.

[3 marks]

10

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0 3 Table 1 shows the running mean of moth catches as a student increased the number of nights of light-trapping at one site.

Table 1

Number of nights	**1**	**2**	**4**	**6**	**8**	**10**	**12**	**14**
Running mean catch per night	31	18	24	21	22	22	21	22

0 3 . 1 Use Table 1 to suggest the minimum number of nights needed to obtain a reliable mean. Explain your answer.

[2 marks]

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0 3 . 2 Explain why moth surveys must be carried out at standardised times and weather conditions.

[3 marks]

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0 3 . 3 Distinguish between **accuracy** and **precision** of measurements.

[2 marks]

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0 3 . 4 Explain the importance of a null hypothesis and standardised techniques in a scientific investigation.

[3 marks]

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0 4 . 1 State what is meant by **hybrid vigour**.

[1 mark]

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0 4 . 2 Explain how cross-breeding Zebu and Ayrshire cattle produced a breed suited to hot climates.

[2 marks]

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0 4 . 3 Give **two** potential applications of cloning to livestock farming.

[2 marks]

1 .....

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2 .....

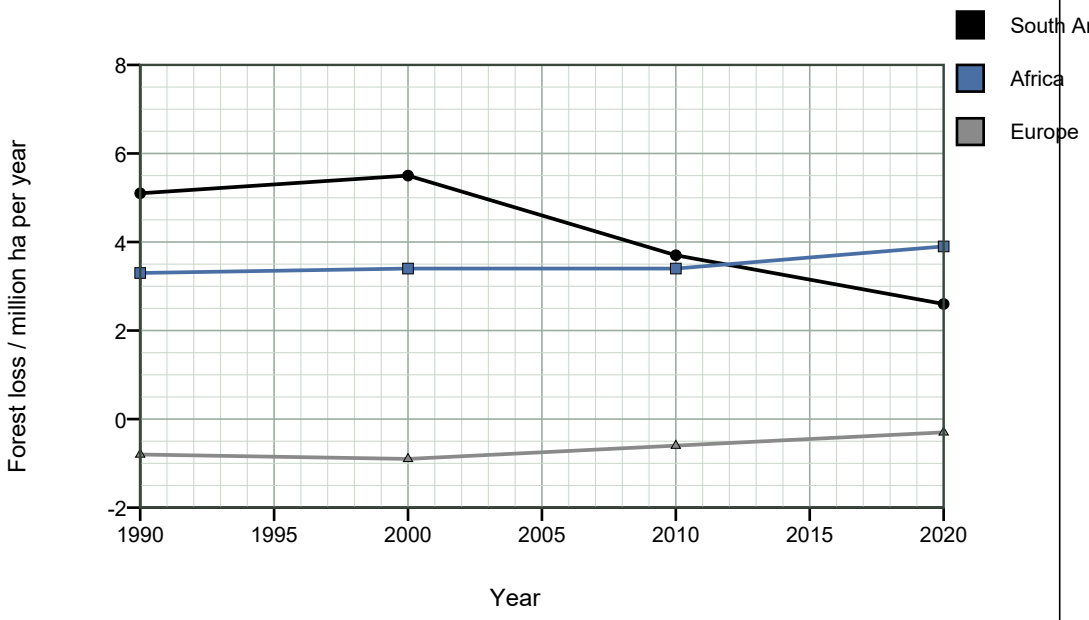
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0 5 Figure 1 shows the area of forest lost each year in three regions between 1990 and 2020.

Figure 1



0 5 . 1 Describe the trends shown in Figure 1.

[3 marks]

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0 5 . 2 Explain the main causes of forest loss in South America and Africa.

[4 marks]

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0 5 . 3 Explain why roads and railways cause more forest loss than the area they occupy.

[2 marks]

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0 5 . 4 Explain why softwood from Scandinavia and Canada is generally produced more sustainably than tropical hardwood.

[1 mark]

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0 6 A national conservation body must decide how to divide a limited budget between species.

0 6 . 1 Describe **three** criteria the IUCN uses when deciding the Red List category of a species.

[3 marks]

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0 6 . 2 A population of lemurs has fallen from 12 000 to 2 100 mature individuals. Calculate the percentage decline.

[2 marks]

Answer [ ] %

0 6 . 3 Explain **three** reasons, other than its Red List category, why one species may be given a higher conservation priority than another.

[3 marks]

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0 6 . 4 Explain the difference between CITES Appendix I and Appendix II, and explain why controlling trade can protect a species more effectively than habitat protection alone.

[3 marks]

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0 6 . 5 Describe **four** ways that UK legislation and designated areas protect habitats.

[4 marks]

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0 7 . 1 Give **two** wild species with potential as new food species and, for each, one characteristic that makes it useful.

[2 marks]

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2 .....

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0 7 . 2 Explain why indigenous species may be better for domestication than introduced livestock or crops.

[2 marks]

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0 7 . 3 Explain how species of the spinach family could help to solve a problem caused by irrigation.

[1 mark]

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5

0 8 . 1 Which feature of the Earth prevents most atmospheric gases from escaping into space? Tick **one** box.

[1 mark]

A Its mass ☐

B Its magnetic field ☐

C Its distance from the Sun ☐

D Its speed of rotation ☐

0 8 . 2 Explain how the speed of rotation of the Earth and the tilt of its axis affect the conditions for life.

[2 marks]

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0 8 . 3 Explain why the presence of liquid water on Earth depends on **both** the distance from the Sun **and** atmospheric pressure.

[2 marks]

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0 9 . 1 Explain **three** features of natural systems that make them sustainable.

[3 marks]

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0 9 . 2 Explain why the human use of mineral resources makes their sustainable exploitation much harder than the use of materials in a natural cycle.

[2 marks]

0 9 . 3 Explain **three** principles of the circular economy that should be applied when a new product is designed.

[3 marks]

0 9 . 4 Explain why an activity that is sustainable for a small population may be unsustainable for a large one. Use **one** example.

[3 marks]

0 9 . 5 Explain why urban populations are less likely than rural communities to notice that a resource is being over-exploited, and give **two** consequences of this.

[4 marks]

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10.1 Discuss the strategies that can be used to increase the sustainability of agriculture.

[9 marks]

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Extra space

1 0 . 2 State what is meant by **permaculture**.

[1 mark]

10

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Write an essay on one of the following topics.

11 . 1 Fishing is the last large-scale human hunting activity. Discuss the environmental impacts of fishing and evaluate the methods used to make the exploitation of fish populations sustainable.

[25 marks]

OR

11 . 2 Explain how the physical features of the Earth and the activities of living organisms have combined to create and maintain conditions suitable for life.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS